IT4090 Cloud Computing · Referral sheet

Lectures 1 to 12 · Exam 16 Nov 2026 · Q1 basics (25) · Q2 case study (25) · Q3 migration + cloud native (25) · Q4 microservices, containers, IaC (25) · 5 MCQs in each question

L1 Introduction

NIST definition

Model for ubiquitous, convenient, on-demand network access to a shared pool of configurable resources that can be rapidly provisioned and released with minimal management effort.

5 characteristics

On-demand self-service · Broad network access · Resource pooling · Rapid elasticity · Measured service

Why cloud

Agility (time to market) · Customer experience (always available) · Cost (pay as you go, **CAPEX → OPEX**) · focus on core business.

	Provider manages	Examples
IaaS	hardware + virtualization; you do OS, apps	EC2, Azure VM, S3
PaaS	+ platform; you do app	Elastic Beanstalk, App Service, RDS
SaaS	everything	Gmail, M365, Salesforce

Deployment	Key points
Public	cheap, scalable, multi-tenant, less control
Private	one org, secure, costly
Hybrid	both, portable, flexible
Community	shared concerns, e.g. government

NIST actors: **consumer, provider, auditor, broker, carrier**. Top CSPs: AWS ~33%, Azure ~20%, GCP ~7%. Cloud = business + delivery model; tech = **virtualization**.

L2 Core services and infrastructure

Region → several **AZs** (isolated, fast private links) → **data centers** (redundant power and network).

Choose Region by: **legal/data governance · latency · services available · cost**.

Core services: **compute, network, storage**. Pay: use-based, **reserve** (RI) for less, **volume** discounts.

Shared responsibility: CSP = security of the cloud; customer = security in the cloud (data, IAM, OS on IaaS, firewall, encryption).

Quotas are per Region. **Soft** limit = can raise; **hard** = cannot.

L3 Compute and storage

VM: guest OS + hypervisor (Type 1 bare metal, Type 2 hosted). **Container:** shares host kernel, light, seconds. Software defined compute = virtualized, pooled.

EC2 = IaaS, AMI, billed per second. Azure VM per minute.

Types: **general · compute · memory · storage · accelerated (GPU) · burstable**. Tenancy: shared · dedicated instance · dedicated host (licences).

Purchase	Key
On-demand	no commitment
Reserved	1 to 3 yr, discount
Spot	cheapest, can be stopped
Savings plan	$/hour commit, flexible

Lifecycle: pending → running → stop / hibernate / **terminate** (permanent). Placement: **cluster** (low latency), **partition** (big distributed), **spread** (max availability). Azure: proximity group.

CAP: pick 2 of Consistency, Availability, Partition tolerance. In a partition: cancel (C) or continue (A).

	Block	File	Object
Data	blocks, IOPS	folders, NFS/SMB	data+metadata+ID, HTTPS
AWS	EBS	EFS	S3
Azure	Disks	Files	Blob

Object features: encryption (client/server; service, customer managed or provided keys), versioning, immutability (retention, legal hold), web hosting. Tiers: **Standard/Hot · Intelligent · IA/Cold · Glacier/Archive**.

L4 Networking

SDN separates control from forwarding. **NFV** runs network functions as software (VNF; CNF in containers).

VPC/VNet: isolated, CIDR, subnets; peer only if no overlap. AWS public subnet has IGW route; private does not. Sizes AWS /16–/28, Azure /2–/29.

Private: 10/8, 172.16/12, 192.168/16. **5 reserved IPs**: network, router, DNS, future, broadcast. **Usable = 2^(32−prefix) − 5**. /24=251, /25=123, /26=59, /27=27, /28=11. First usable = network + 4.

	Applies to	State
Security group	instance	stateful
NACL	subnet	stateless
Azure NSG	subnet+NIC	stateful

IGW internet. **NAT gateway** (in public subnet) outbound only. **VPN**: site-to-site, point-to-site (AWS Client VPN). **Transit gateway / Virtual WAN** hub and spoke. **Peering** not transitive. **Bastion** in public subnet.

Hybrid: VPN or **Direct Connect / ExpressRoute** (private line). Private Link / endpoint = PaaS without internet. **Flow logs**: IP traffic, no latency impact.

L5 HA and DR

Downtime/yr: 99% 3.65 d · 99.9% 8.77 h · 99.99% 52.6 min · 99.999% 5.26 min.

HA: remove SPOF · reliable crossover · detect failure. Redundancy **passive** (spare capacity) / **active** (auto reconfigure). **Failover** (heartbeat) / **failback**. Replication active/passive. Cluster active-active / active-passive.

Load balancing: round robin · least connections · least time · IP hash; session persistence.

BCP (impact + threat analysis) ⊃ **DR** (team, risk, critical assets, backup, testing). Backup: **full · incremental · differential** · CDP. DR tiers 0–7 (7 = near zero loss).

RTO time to restore · **RPO** acceptable data loss · **RCO** consistency.

L6 Security and monitoring

PLOP: only the privileges needed. **IAM:** identify, authenticate, authorize. Components: **logical organization · users · groups · policies · roles** (temporary credentials).

MFA factors: have · know · are · where. **Federation** links identities; **SSO** one login. Enterprise vs web identity federation.

Symmetric one key (fast). **Asymmetric** public + private. **Hash** one-way. Data **in use / in transit (TLS) / at rest**. Client-side vs server-side encryption.

Risks: poor access mgmt · breach · misconfiguration · insecure APIs · hijacking · no visibility · DoS/DDoS.

Monitoring: CloudWatch, Azure Monitor. **Metrics** numbers · **events** what happened · **logs** history · **alerts**: threshold, anomaly, heartbeat → notify or auto action. Software defined security = policy-driven, automated.

L7 Database and analytics

SQL: tables, schema, **ACID**. **NoSQL:** flexible. Types: **key-value** (Redis, DynamoDB: session, cart) · **document** (MongoDB) · **columnar** (Cassandra) · **graph** (Neptune: relationships).

	OLTP	OLAP
Data	current, transactions	historical, analysis
Query	small, ms	large, seconds–hours
Design	normalized	denormalized

Cache: memory, less DB load. **Redis:** persistent, many structures, replicas. **Memcached:** not persistent, multithreaded.

Warehouse: processed, structured, BI, ETL (transform before load). **Lake:** raw, flat, data scientists, ELT. **Real-time** = stream processing (Kinesis).

L8 Cloud migration

Phases: **opportunity evaluation → discovery and planning → design, migrate, validate → operate** (DevOps, NoOps, FinOps, SecOps). Tools: AWS Discovery, Azure Migrate. Start with simplest apps; parallel run; cutover.

6 Rs	Meaning
Rehost	lift and shift, no change
Re-platform	lift, tinker, shift (DB to RDS)
Repurchase	move to SaaS (CRM to Salesforce)
Refactor	re-architect, cloud native, costliest
Retire	switch off
Retain	keep for now

Gartner 5 Rs: rehost, refactor, revise, rebuild, replace. Azure 4 Rs: rehost, refactor, re-architect, rebuild. Challenges: **cost · resistance to adapt · skills shortage** (with fixes).

L9 Cloud native

Monolith: single unit; simple but hard to change/scale. **Microservices:** small independent services, APIs, scale each, but distributed complexity. **Serverless/FaaS:** functions on events, pay per run (Lambda).

12-factor: codebase · dependencies · config (env vars) · backing services · build-release-run · processes (stateless) · port binding · concurrency · disposability · dev/prod parity · logs · admin processes.

Tools: containers, microservices, **service mesh**, orchestration. DevOps · SRE · NoOps. **CI** merge + auto test · **Delivery** manual approval · **Deployment** automatic. Modernization: rehost → container → add microservices → rearchitect → rebuild.

L10 IaC

Manual risks: no repeatability, version control, audit trail; inconsistency. **IaC** = infrastructure defined in template files. **Declarative** (end state) vs **imperative** (steps).

Tool types: **config mgmt** (Ansible, Puppet, Chef) · **templating** (Docker, Packer) · **orchestration** (Kubernetes) · **provisioning** (Terraform, CloudFormation).

Benefits: consistency, speed, version control, reuse, easy clean-up. Best practice: version control, test, modular. **Terraform:** write → plan → apply. **CloudFormation:** template (JSON/YAML) → stack, AWS only.

L11 Containers and Docker

Container = app + dependencies, shares host kernel. Objects: **image** (read-only blueprint) · **container** (running) · **Dockerfile** · **registry** (Docker Hub, ECR, ACR). Parts: **daemon** + **client**.

```
docker pull IMG      docker create --name N IMG
docker start N       docker run IMG (= pull+create+start)
docker ps / ps -a    docker logs N     docker attach N
docker exec -it N cmd   docker pause|unpause N
docker stop N (SIGTERM)  docker kill N (SIGKILL)
docker restart N    docker rm N    docker rmi IMG    docker system prune
```

L12 Orchestration

Automates deploy, manage, scale, network of containers. Cluster = **worker nodes** + **control plane**. Cycle: **provisioning → deployment → lifecycle management**. Benefits: automation, self-healing, efficiency, portability, DevOps fit. Tools: **Kubernetes** (EKS/AKS/GKE), Docker Swarm, OpenShift. K8s: cluster, node, pod, service.

Exam tips: no reading time · MCQs first, fast · Q2 case study: use the scenario's facts, pick service + deployment model, cover scalability, security, availability, cost · compare = table · skip Lecture 13 topics (well-architected, multi-cloud, innovation barriers).